Data Assimilation:
From the Kalman Filter to Modern Approaches

Mao Zhiyuan
School of Mathematical Sciences, Peking University

1 Introduction

Data assimilation (DA) lies at the core of modern data science and computational modeling. It refers to a class of methods that combine prior model predictions with observational data in order to obtain a more accurate estimate of the state of a system (Asch et al., 2016; Carrassi et al., 2018).

Weather forecasting provides a typical example. Suppose we are interested in a set of atmospheric variables, such as temperature, pressure, humidity, and wind fields, across the entire three-dimensional atmosphere. In practice, however, observations are limited and cannot be directly used to describe the full system state, due to several sources of difficulty:

- **Spatial inconsistency:** observations are unavailable in certain regions;
- **Temporal inconsistency:** observations are unavailable at certain times;
- **Measurement errors:** observational data are affected by noise and instrument uncertainty;
- **Proxy errors:** variables of interest may not be observed directly, and only related proxy variables are available;
- **Model errors:** numerical models are imperfect and may introduce systematic or random errors.

As a result, neither observations nor model predictions alone are sufficient. Data assimilation serves as a framework for balancing the two sources of information by weighting them according to their respective uncertainties. In this way, it produces an improved estimate that is generally more reliable than either the raw observations or the model forecast alone.

The rest of this report is organized as follows. Section 2 presents the general framework and formulation of state-space models. Section 3 discusses the canonical Kalman filter, including its key properties and limitations. Section 4 provides a brief introduction to modern approaches in data assimilation. Finally, Section 5 concludes the report. A detailed derivation of the Kalman filter is collected in Appendix A.

2 General Framework: State-Space Model

The state-space model provides a general framework that formalizes the intuition described above. For a system of interest, there exists a set of time-varying latent variables, denoted by x_t, which cannot be observed or measured directly. These hidden states generate another set of observable variables, denoted by y_t. A canonical state-space model consists of two components: the **state**

transition equation, which describes the evolution of the hidden state over time based on prior knowledge of the system dynamics, and the **observation equation**, which specifies how observed data are generated from the unobservable state variables.

In its most general form, a state-space model can be written as

$$x_t = f_t(x_{t-1}, u_t, w_t),$$

$$y_t = h_t(x_t, u_t, v_t),$$

where x_t represents the hidden state of the system at time step t, namely the unobservable variables of interest; y_t denotes the observed data at time step t; u_t is an external input or control variable; w_t is the noise term in the state transition model; and v_t is the noise term in the observation model. The functions f_t and h_t are generally time-varying functions that describe the state transition process and the observation-generating process, respectively.

In many applications, the model is specified with additive noise:

$$x_t = f_t(x_{t-1}, u_t) + w_t,$$

$$y_t = h_t(x_t, u_t) + v_t.$$

The state transition equation determines the transition density $p(x_t \mid x_{t-1})$, while the observation equation determines the likelihood $p(y_t \mid x_t)$.

The central goal of data assimilation is to characterize the posterior distribution of the hidden state x_t given all observations up to time t, denoted by $y_{1:t}$. Since the model is Markovian, this posterior distribution can be computed recursively through two steps.

First, the **time update**, or prediction step, propagates the posterior distribution from time $t-1$ to time t:

$$p(x_t \mid y_{1:t-1}) = \int p(x_t \mid x_{t-1}) p(x_{t-1} \mid y_{1:t-1}) \, \mathrm{d}x_{t-1}.$$

Second, the **measurement update**, or correction step, incorporates the new observation y_t using Bayes' formula:

$$p(x_t \mid y_{1:t}) = \frac{p(y_t \mid x_t) p(x_t \mid y_{1:t-1})}{\int p(y_t \mid x_t) p(x_t \mid y_{1:t-1}) \, \mathrm{d}x_t} \propto p(y_t \mid x_t) p(x_t \mid y_{1:t-1}).$$

These two recursive equations provide the general Bayesian formulation for combining model predictions with observational data. However, in the absence of linearity and Gaussian error assumptions, the integrals involved in the prediction and correction steps typically do not admit closed-form solutions. This makes practical data assimilation challenging and motivates the development of approximate methods such as the extended Kalman filter, ensemble Kalman filter, particle filter, and variational approaches.

3 The Kalman Filter

As discussed in the previous section, a general state-space model does not necessarily admit a closed-form solution. When both the state transition and observation equations are linear and all random errors are Gaussian, however, the prediction and correction distributions remain Gaussian at every time step. Consequently, the entire posterior distribution can be characterized by recursively updating only its mean and variance. This recursive procedure is known as the **Kalman filter** (Kalman, 1960; Anderson and Moore, 1979).

3.1 Linear Gaussian Model

Consider the following linear state-space model:

$$\boldsymbol{x}_t = \boldsymbol{A}_t \boldsymbol{x}_{t-1} + \boldsymbol{B}_t \boldsymbol{u}_t + \boldsymbol{w}_t, \qquad \boldsymbol{y}_t = \boldsymbol{H}_t \boldsymbol{x}_t + \boldsymbol{v}_t,$$

where $\boldsymbol{x}_t \in \mathbb{R}^n$ is the hidden state, $\boldsymbol{y}_t \in \mathbb{R}^m$ is the observation, and $\boldsymbol{u}_t$ is a known external input. The matrices $\boldsymbol{A}_t$, $\boldsymbol{B}_t$, and $\boldsymbol{H}_t$ are assumed known. The state and observation errors satisfy

$$\boldsymbol{w}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_t), \qquad \boldsymbol{v}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_t), \qquad \boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{m}_0, \boldsymbol{P}_0).$$

It is assumed that $\boldsymbol{x}_0$, $\boldsymbol{w}_t$, and $\boldsymbol{v}_t$ are mutually independent, and that both noise sequences are independent across time.

Let $\boldsymbol{y}_{1:t} = (\boldsymbol{y}_1, \ldots, \boldsymbol{y}_t)$ denote all observations available up to time t. The filtering problem is to determine $p(\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t})$. We write the filtered mean and variance as

$$\hat{\boldsymbol{x}}_t = \mathbb{E}[\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t}], \qquad \boldsymbol{P}_t = \mathrm{Var}(\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t}),$$

and the one-step-ahead prediction quantities as

$$\hat{\boldsymbol{x}}_t^- = \mathbb{E}[\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t-1}], \qquad \boldsymbol{P}_t^- = \mathrm{Var}(\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t-1}).$$

The superscript $-$ indicates that the observation $\boldsymbol{y}_t$ has not yet been incorporated.

The key fact underlying the Kalman filter is the closure of Gaussian distributions under linear transformations, marginalization, and conditioning. If the filtering distribution at time $t-1$ is Gaussian, then applying the linear state transition equation produces a Gaussian prediction distribution, and combining this prediction with the linear Gaussian likelihood again gives a Gaussian posterior. Starting from the Gaussian initial distribution, it follows by induction that

$$\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t} \sim \mathcal{N}(\hat{\boldsymbol{x}}_t, \boldsymbol{P}_t)$$

for every t. Hence, rather than approximating an arbitrary probability density, it is sufficient to track only $\hat{\boldsymbol{x}}_t$ and $\boldsymbol{P}_t$.

3.2 The Kalman Filter Recursion

Under the linear Gaussian model, the filtering distribution stays Gaussian at every step, and the mean and variance are propagated by the two-step recursion stated below. The full derivation, which relies only on the linear-transformation and conditioning properties of Gaussian distributions, is collected in Appendix A; here we record only the resulting equations.

Prediction step. Given the filtered estimate $(\hat{\boldsymbol{x}}_{t-1}, \boldsymbol{P}_{t-1})$ at time $t-1$, the predicted mean and variance are

$$\boxed{\hat{\boldsymbol{x}}_t^- = \boldsymbol{A}_t \hat{\boldsymbol{x}}_{t-1} + \boldsymbol{B}_t \boldsymbol{u}_t} \qquad \boxed{\boldsymbol{P}_t^- = \boldsymbol{A}_t \boldsymbol{P}_{t-1} \boldsymbol{A}_t^\top + \boldsymbol{Q}_t}$$

The prediction mean propagates the previous estimate through the dynamical model, while the prediction variance combines the propagated prior uncertainty with the additional uncertainty introduced by the process noise $\boldsymbol{Q}_t$.

Observation update. To incorporate the new observation $\boldsymbol{y}_t$, define the **innovation** (the difference between the observation and its one-step prediction) and its variance,

$$\tilde{\boldsymbol{y}}_t = \boldsymbol{y}_t - \boldsymbol{H}_t \hat{\boldsymbol{x}}_t^-, \qquad \boldsymbol{S}_t = \boldsymbol{H}_t \boldsymbol{P}_t^- \boldsymbol{H}_t^\top + \boldsymbol{R}_t.$$

The **Kalman gain** and the filtered mean and variance are then

$$\boxed{\boldsymbol{K}_t = \boldsymbol{P}_t^- \boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1}} \qquad \boxed{\hat{\boldsymbol{x}}_t = \hat{\boldsymbol{x}}_t^- + \boldsymbol{K}_t \tilde{\boldsymbol{y}}_t} \qquad \boxed{\boldsymbol{P}_t = (\boldsymbol{I} - \boldsymbol{K}_t \boldsymbol{H}_t)\, \boldsymbol{P}_t^-}$$

The Kalman gain determines how strongly the estimate responds to the innovation. When the observation uncertainty $\boldsymbol{R}_t$ is small relative to the prediction uncertainty, the gain is large and the observation receives more weight; when the observations are highly uncertain, the gain is small and the updated estimate remains close to the model forecast.

3.3 Other Interpretations and Properties

The recursion above admits several complementary interpretations, and the filtered quantities carry additional structure that is useful in practice. In this part we introduce the main results which can be referred to (Anderson and Moore, 1979; Simon, 2006; Särkkä, 2013).

Minimum-variance and BLUE interpretation. The same update can be derived without any Gaussian assumption, by choosing the gain that minimizes the trace of the posterior error variance among all linear estimators. The minimizer is exactly the Kalman gain $\boldsymbol{K}_t$, so the Kalman update is the minimum mean-square linear estimator and, when the prediction is unbiased, the best linear unbiased estimator (BLUE). This derivation also yields the numerically more robust **Joseph form** of the variance update,

$$\boldsymbol{P}_t = (\boldsymbol{I} - \boldsymbol{K}_t \boldsymbol{H}_t)\, \boldsymbol{P}_t^- \,(\boldsymbol{I} - \boldsymbol{K}_t \boldsymbol{H}_t)^\top + \boldsymbol{K}_t \boldsymbol{R}_t \boldsymbol{K}_t^\top,$$

which is algebraically equivalent to $\boldsymbol{P}_t = (\boldsymbol{I} - \boldsymbol{K}_t \boldsymbol{H}_t)\boldsymbol{P}_t^-$ but explicitly preserves symmetry and positive semidefiniteness. Under the full Gaussian model, the conditional mean is linear in the observation, so the Kalman estimate is in addition the unrestricted minimum mean-square error (MMSE) estimator.

Innovation whiteness. Under a correctly specified model, the innovation sequence $\tilde{\boldsymbol{y}}_t$ has zero conditional mean, conditional variance $\boldsymbol{S}_t$, and is uncorrelated across time (i.e. white). The normalized innovations $\boldsymbol{S}_t^{-1/2}\tilde{\boldsymbol{y}}_t$ should therefore behave approximately as independent standard Gaussian vectors. Persistent temporal correlation, a nonzero mean, or an empirical variance far from $\boldsymbol{S}_t$ signals misspecified dynamics, biased observations, or poor choices of $\boldsymbol{Q}_t$ and $\boldsymbol{R}_t$, which makes innovation analysis a natural model diagnostic.

Information-filter form. Writing the filter in terms of the inverse variance, or *information matrix* $\boldsymbol{\Lambda}_t = \boldsymbol{P}_t^{-1}$ and information vector $\boldsymbol{\eta}_t = \boldsymbol{\Lambda}_t \hat{\boldsymbol{x}}_t$, the measurement update becomes purely additive,

$$\boldsymbol{\Lambda}_t = \boldsymbol{\Lambda}_t^- + \boldsymbol{H}_t^\top \boldsymbol{R}_t^{-1} \boldsymbol{H}_t, \qquad \boldsymbol{\eta}_t = \boldsymbol{\eta}_t^- + \boldsymbol{H}_t^\top \boldsymbol{R}_t^{-1} \boldsymbol{y}_t,$$

so each independent observation contributes an additive term. This form is convenient when fusing many independent data sources or when the information matrices are sparse.

Riccati recursion. Combining the prediction and update variance equations yields a nonlinear matrix recursion for $\boldsymbol{P}_t$, the discrete Riccati recursion. Notably, it does not depend on the realized observations, so the sequence of variances and Kalman gains can be precomputed offline once the model matrices and the initial variance are specified.

3.4 Limitations and Practical Challenges

The Kalman filter is exact and optimal, but only under assumptions that are frequently violated in real applications. Its limitations fall along three axes, which directly motivate the methods of Section 4.

- **Linearity and Gaussianity.** The recursion assumes linear dynamics and observations with Gaussian noise. When f_t or h_t is nonlinear, or when the posterior is skewed, heavy-tailed, or multimodal, a mean and variance can no longer describe it, and the Kalman update may be badly biased.
- **High dimensionality.** The filter stores and propagates the full $n \times n$ variance $\boldsymbol{P}_t$. In geophysical systems the state dimension n can reach 10^7 or more, making explicit storage and the $\boldsymbol{A}_t \boldsymbol{P}_{t-1} \boldsymbol{A}_t^\top$ propagation computationally infeasible.
- **Reliance on a known model.** The matrices $\boldsymbol{A}_t, \boldsymbol{H}_t$ and the noise variances $\boldsymbol{Q}_t, \boldsymbol{R}_t$ are assumed given. In practice the dynamics are imperfect and the variances must be tuned; misspecification can cause the estimated uncertainty to collapse and the filter to diverge.

These three difficulties—nonlinear and non-Gaussian uncertainty, high dimensionality, and the need to learn or correct the model from data—are precisely what the modern approaches in Section 4 are designed to address. Each relaxes one of the assumptions above while retaining the basic Bayesian idea of weighting a forecast against observations by their relative uncertainty.

4 Beyond the Kalman Filter: Modern Approaches

The Kalman filter developed in Section 3 is exact for linear Gaussian state-space models. Its practical success comes from the fact that the full posterior distribution can be summarized by the mean $\hat{\boldsymbol{x}}_t$ and variance $\boldsymbol{P}_t$. Modern data assimilation methods can be understood as attempts to relax the assumptions behind this construction. In particular, they address three major difficulties: high dimensionality, non-Gaussian uncertainty, and the need to learn parts of the dynamical model from data.

4.1 High-Dimensional Systems: Ensemble Kalman Filtering

In many geophysical applications, the state dimension n is extremely large. For example, a global atmospheric state may contain temperature, pressure, wind, humidity, and other variables on a three-dimensional grid, so n can easily reach 10^7 or more. In this regime, storing and propagating the full variance matrix $\boldsymbol{P}_t \in \mathbb{R}^{n \times n}$ is impossible. The ensemble Kalman filter, or EnKF, keeps the basic Kalman update structure but represents uncertainty by a finite ensemble of model states (Evensen, 1994, 2009).

Suppose that the prediction ensemble at time t is $\{\boldsymbol{x}_t^{(i),-}\}_{i=1}^N$, where N is the ensemble size. The prediction mean and variance are approximated by the sample quantities

$$\bar{\boldsymbol{x}}_t^- = \frac{1}{N}\sum_{i=1}^N \boldsymbol{x}_t^{(i),-}, \qquad \boldsymbol{P}_{t,\mathrm{ens}}^- = \frac{1}{N-1}\sum_{i=1}^N \left(\boldsymbol{x}_t^{(i),-} - \bar{\boldsymbol{x}}_t^-\right)\left(\boldsymbol{x}_t^{(i),-} - \bar{\boldsymbol{x}}_t^-\right)^\top.$$

The Kalman gain is then approximated by replacing $\boldsymbol{P}_t^-$ with $\boldsymbol{P}_{t,\mathrm{ens}}^-$:

$$\boldsymbol{K}_{t,\mathrm{ens}} = \boldsymbol{P}_{t,\mathrm{ens}}^- \boldsymbol{H}_t^\top \left(\boldsymbol{H}_t \boldsymbol{P}_{t,\mathrm{ens}}^- \boldsymbol{H}_t^\top + \boldsymbol{R}_t\right)^{-1}.$$

The ensemble members are updated by a Kalman-type correction using the innovation $\boldsymbol{y}_t - \boldsymbol{H}_t \boldsymbol{x}_t^{(i),-}$. Thus, the EnKF preserves the main idea of the Kalman filter: the model forecast provides a prior, and the observation corrects it according to relative uncertainty. The crucial difference is that the variance is no longer stored explicitly; it is estimated from the ensemble.

In practice, N is usually much smaller than n. Therefore, the sample variance is low-rank and contains sampling noise, especially in long-range correlations. Two important regularization techniques are commonly used. The first is **localization**, which damps spurious correlations between distant state variables. If $\boldsymbol{C}$ is a prescribed correlation taper matrix, localization replaces the empirical variance by

$$\boldsymbol{P}_{t,\mathrm{loc}}^- = \boldsymbol{C} \circ \boldsymbol{P}_{t,\mathrm{ens}}^-,$$

where $\circ$ denotes the elementwise product. The second is **inflation**, which compensates for the tendency of small ensembles to underestimate uncertainty. A simple multiplicative inflation takes the form

$$\boldsymbol{P}_{t,\mathrm{inf}}^- = \lambda \boldsymbol{P}_{t,\mathrm{loc}}^-, \qquad \lambda > 1.$$

With localization and inflation, the Kalman filtering idea becomes feasible for very high-dimensional systems. This is one reason why ensemble-based methods are central in modern weather and ocean data assimilation. Their limitation is that the update is still essentially Gaussian and linear in the observation; strong nonlinearity or multimodal uncertainty may not be represented accurately by only the ensemble mean and variance.

4.2 Non-Gaussian Problems: Particle Filtering

A different way to go beyond the Kalman filter is to avoid the Gaussian approximation altogether. Particle filters represent the filtering distribution directly by weighted samples (Gordon et al., 1993; Doucet et al., 2001):

$$p(\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t}) \approx \sum_{i=1}^N w_t^{(i)} \delta\left(\boldsymbol{x}_t - \boldsymbol{x}_t^{(i)}\right),$$

where $\boldsymbol{x}_t^{(i)}$ is the i-th particle, $w_t^{(i)}$ is its normalized weight, and $\delta(\cdot)$ is the Dirac measure. Unlike the Kalman filter or EnKF, this approximation can represent skewness, heavy tails, and multiple modes.

In the bootstrap particle filter, particles are first propagated through the state transition model. The observation is then used to update the weights:

$$\tilde{w}_t^{(i)} \propto w_{t-1}^{(i)} p(\boldsymbol{y}_t \mid \boldsymbol{x}_t^{(i)}), \qquad w_t^{(i)} = \frac{\tilde{w}_t^{(i)}}{\sum_{j=1}^N \tilde{w}_t^{(j)}}.$$

Intuitively, particles that better explain the observation receive larger weights. When the weights become too uneven, resampling is applied to duplicate high-weight particles and remove low-weight particles. A common diagnostic is the effective sample size

$$N_{\text{eff}} = \frac{1}{\sum_{i=1}^{N} \left(w_t^{(i)}\right)^2},$$

where a small N_{eff} indicates that only a few particles dominate the approximation.

The strength of particle filters is their generality. They can approximate arbitrary posterior distributions and do not require linear dynamics or Gaussian noise. This makes them conceptually closer to the full Bayesian recursion in Section 2. Their main weakness is **weight degeneracy**. In high-dimensional systems, the likelihood can vary enormously across particles, so almost all weights collapse to zero unless the number of particles is extremely large. This is a manifestation of the curse of dimensionality. Consequently, particle filters are powerful for low- or moderate-dimensional nonlinear systems, but naive particle filtering is usually infeasible for operational-scale atmospheric data assimilation.

4.3 Machine Learning and Generative Data Assimilation

A more recent direction is to use machine learning to learn either the forecast model, the prior distribution, or the full assimilation map from data. These methods do not simply modify the Kalman filter update; instead, they attempt to replace parts of the traditional numerical pipeline.

One use of machine learning is to build a fast surrogate for the forecast model. Instead of applying a hand-designed numerical solver, one trains a neural network $\mathcal{M}_\theta$ so that

$$\boldsymbol{x}_{t+\Delta} \approx \mathcal{M}_\theta(\boldsymbol{x}_t, \boldsymbol{x}_{t-\Delta}).$$

GraphCast is an example of this idea in weather forecasting: it learns a data-driven global forecast model from reanalysis data (Lam et al., 2023). In this setting, data assimilation is still needed to provide an accurate initial condition, but the expensive forward model can be replaced or accelerated by the learned model.

A more ambitious approach is end-to-end learning. Instead of separating observation processing, data assimilation, forecasting, and post-processing into different components, an end-to-end model learns a direct mapping from observations to forecasts. Schematically, such a system may be written as

$$\hat{\boldsymbol{x}}_0 = E_\theta(\boldsymbol{y}_0), \qquad \hat{\boldsymbol{x}}_{t+\Delta} = \mathcal{M}_\theta(\hat{\boldsymbol{x}}_t), \qquad \hat{\boldsymbol{y}}_t = D_\theta(\hat{\boldsymbol{x}}_t),$$

where E_θ encodes observations into an estimated initial state, $\mathcal{M}_\theta$ advances the state, and D_θ maps the state to user-level forecast quantities. Aardvark Weather is an example of this direction (Allen et al., 2025). Its goal is not merely to replace the forecast model, but to learn a complete observation-to-forecast pipeline.

Another active direction is **score-based** or **diffusion-based** data assimilation (Song and Ermon, 2019; Rozet and Louppe, 2023). The central object is the score function $\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})$, the gradient of the log-density of the prior. A score network s_θ is trained so that $s_\theta(\boldsymbol{x}) \approx \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})$. After training, observations can be incorporated by Bayes' rule. Since $p(\boldsymbol{x} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{x})p(\boldsymbol{x})$, the posterior score satisfies

$$\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x} \mid \boldsymbol{y}) = \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} \log p(\boldsymbol{y} \mid \boldsymbol{x}).$$

For the linear Gaussian observation model $\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{v}$ with $\boldsymbol{v} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R})$, this becomes

$$\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x} \mid \boldsymbol{y}) \approx s_\theta(\boldsymbol{x}) + \boldsymbol{H}^\top \boldsymbol{R}^{-1}(\boldsymbol{y} - \boldsymbol{H}\boldsymbol{x}).$$

The posterior can then be sampled using Langevin dynamics or a reverse diffusion process, for example

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \frac{\epsilon}{2}\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}^k \mid \boldsymbol{y}) + \sqrt{\epsilon}\boldsymbol{\xi}^k, \qquad \boldsymbol{\xi}^k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}).$$

The advantage of this framework is that it aims to sample from the full posterior distribution, rather than only estimating a mean and variance. It is therefore potentially useful for strongly nonlinear and non-Gaussian problems, and it can be applied to entire trajectories $\boldsymbol{x}_{0:T}$, not only to a single state $\boldsymbol{x}_t$. The main challenges are the need for large training datasets, the difficulty of ensuring physical consistency, and the calibration of posterior uncertainty.

Overall, modern data assimilation does not abandon the Kalman filter idea; it generalizes it in different directions. EnKF methods keep the Kalman update but make variance estimation feasible in high dimensions. Particle filters keep the full Bayesian posterior but suffer from degeneracy in large systems. Machine learning and generative methods attempt to learn forecast dynamics or posterior distributions directly from data. In practice, the most promising systems are often hybrid: they combine physical models, ensemble uncertainty quantification, and learned components.

5 Conclusion

This report has reviewed data assimilation from the classical Kalman filter to several modern extensions. The central idea throughout is to combine model predictions with observations in a statistically principled way, so that the posterior estimate of the hidden state $\boldsymbol{x}_t$ is more reliable than either source of information alone.

The Kalman filter provides an exact and elegant solution under linear Gaussian assumptions, where the posterior distribution is completely characterized by its mean $\hat{\boldsymbol{x}}_t$ and variance $\boldsymbol{P}_t$. These guarantees, however, rely on linearity, Gaussianity, and the ability to store and propagate the full variance matrix—assumptions that break down in the nonlinear, non-Gaussian, and very high-dimensional settings encountered in practice.

Modern methods relax these different limitations. Ensemble Kalman filters make variance-based assimilation feasible for very high-dimensional systems; particle filters provide a more general Bayesian approximation for nonlinear and non-Gaussian problems; and machine-learning or generative approaches attempt to learn forecast dynamics, assimilation maps, or posterior distributions directly from data. In practice, future data assimilation systems are likely to be hybrid, combining physical models, statistical uncertainty quantification, and learned components.

References

Anna Allen, Stratis Markou, Will Tebbutt, James Requeima, Wessel P. Bruinsma, Tom R. Andersson, Michael Herzog, Nicholas D. Lane, Matthew Chantry, J. Scott Hosking, and Richard E. Turner. End-to-end data-driven weather prediction. *Nature*, 641(8065):1172–1179, 2025.

Brian D. O. Anderson and John B. Moore. *Optimal Filtering*. Prentice-Hall, Englewood Cliffs, NJ, 1979.

Mark Asch, Marc Bocquet, and Maëlle Nodet. *Data Assimilation: Methods, Algorithms, and Applications*, volume 11 of *Fundamentals of Algorithms*. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2016.

Alberto Carrassi, Marc Bocquet, Laurent Bertino, and Geir Evensen. Data assimilation in the geosciences: An overview of methods, issues, and perspectives. *WIREs Climate Change*, 9(5): e535, 2018.

Arnaud Doucet, Nando de Freitas, and Neil Gordon, editors. *Sequential Monte Carlo Methods in Practice*. Statistics for Engineering and Information Science. Springer, New York, 2001.

Geir Evensen. Sequential data assimilation with a nonlinear quasi-geostrophic model using Monte Carlo methods to forecast error statistics. *Journal of Geophysical Research: Oceans*, 99(C5): 10143–10162, 1994.

Geir Evensen. *Data Assimilation: The Ensemble Kalman Filter*. Springer, Berlin, 2nd edition, 2009.

Neil J. Gordon, David J. Salmond, and Adrian F. M. Smith. Novel approach to nonlinear/non-Gaussian Bayesian state estimation. *IEE Proceedings F (Radar and Signal Processing)*, 140(2): 107–113, 1993.

Rudolph E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, Peter Wirnsberger, Meire Fortunato, Ferran Alet, Suman Ravuri, Timo Ewalds, Zach Eaton-Rosen, Weihua Hu, et al. Learning skillful medium-range global weather forecasting. *Science*, 382(6677):1416–1421, 2023.

François Rozet and Gilles Louppe. Score-based data assimilation. In *Advances in Neural Information Processing Systems*, volume 36, pages 40521–40541, 2023.

Simo Särkkä. *Bayesian Filtering and Smoothing*. Institute of Mathematical Statistics Textbooks. Cambridge University Press, Cambridge, 2013.

Dan Simon. *Optimal State Estimation: Kalman, H_∞, and Nonlinear Approaches*. John Wiley & Sons, Hoboken, NJ, 2006.

Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

A Derivation of the Kalman Filter

This appendix derives the prediction and update equations stated in Section 3.2, using only the linear-transformation and conditioning properties of Gaussian distributions.

A.1 Prediction Step

Suppose the filtering distribution at time $t-1$ is $\boldsymbol{x}_{t-1} \mid \boldsymbol{y}_{1:t-1} \sim \mathcal{N}(\hat{\boldsymbol{x}}_{t-1}, \boldsymbol{P}_{t-1})$. The state transition equation is $\boldsymbol{x}_t = \boldsymbol{A}_t \boldsymbol{x}_{t-1} + \boldsymbol{B}_t \boldsymbol{u}_t + \boldsymbol{w}_t$. Conditional on $\boldsymbol{y}_{1:t-1}$, the input $\boldsymbol{u}_t$ is known and $\boldsymbol{w}_t$ has zero mean, so

$$\begin{aligned}\hat{\boldsymbol{x}}_t^- &= \mathbb{E}[\boldsymbol{A}_t \boldsymbol{x}_{t-1} + \boldsymbol{B}_t \boldsymbol{u}_t + \boldsymbol{w}_t \mid \boldsymbol{y}_{1:t-1}] \\ &= \boldsymbol{A}_t \mathbb{E}[\boldsymbol{x}_{t-1} \mid \boldsymbol{y}_{1:t-1}] + \boldsymbol{B}_t \boldsymbol{u}_t = \boldsymbol{A}_t \hat{\boldsymbol{x}}_{t-1} + \boldsymbol{B}_t \boldsymbol{u}_t.\end{aligned}$$

Let $\boldsymbol{e}_{t-1} = \boldsymbol{x}_{t-1} - \hat{\boldsymbol{x}}_{t-1}$ be the filtering error. The prediction error is

$$\boldsymbol{e}_t^- = \boldsymbol{x}_t - \hat{\boldsymbol{x}}_t^- = \boldsymbol{A}_t \left(\boldsymbol{x}_{t-1} - \hat{\boldsymbol{x}}_{t-1}\right) + \boldsymbol{w}_t = \boldsymbol{A}_t \boldsymbol{e}_{t-1} + \boldsymbol{w}_t.$$

Since $\boldsymbol{e}_{t-1}$ and $\boldsymbol{w}_t$ are independent, their variance is zero, and therefore

$$\begin{aligned}\boldsymbol{P}_t^- &= \mathbb{E}[\boldsymbol{e}_t^- (\boldsymbol{e}_t^-)^\top \mid \boldsymbol{y}_{1:t-1}] \\ &= \mathbb{E}[(\boldsymbol{A}_t \boldsymbol{e}_{t-1} + \boldsymbol{w}_t)(\boldsymbol{A}_t \boldsymbol{e}_{t-1} + \boldsymbol{w}_t)^\top \mid \boldsymbol{y}_{1:t-1}] = \boldsymbol{A}_t \boldsymbol{P}_{t-1} \boldsymbol{A}_t^\top + \boldsymbol{Q}_t.\end{aligned}$$

This establishes the prediction equations $\hat{\boldsymbol{x}}_t^- = \boldsymbol{A}_t \hat{\boldsymbol{x}}_{t-1} + \boldsymbol{B}_t \boldsymbol{u}_t$ and $\boldsymbol{P}_t^- = \boldsymbol{A}_t \boldsymbol{P}_{t-1} \boldsymbol{A}_t^\top + \boldsymbol{Q}_t$, and the prediction distribution is $\boldsymbol{x}_t \mid \boldsymbol{y}_{1:t-1} \sim \mathcal{N}(\hat{\boldsymbol{x}}_t^-, \boldsymbol{P}_t^-)$.

A.2 Observation Update

Conditional on $\boldsymbol{y}_{1:t-1}$, the predicted state satisfies $\boldsymbol{x}_t \sim \mathcal{N}(\hat{\boldsymbol{x}}_t^-, \boldsymbol{P}_t^-)$, and the observation equation is $\boldsymbol{y}_t = \boldsymbol{H}_t \boldsymbol{x}_t + \boldsymbol{v}_t$. Hence the predicted observation has mean $\mathbb{E}[\boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1}] = \boldsymbol{H}_t \hat{\boldsymbol{x}}_t^-$, and the innovation $\tilde{\boldsymbol{y}}_t = \boldsymbol{y}_t - \boldsymbol{H}_t \hat{\boldsymbol{x}}_t^-$ can be written as

$$\tilde{\boldsymbol{y}}_t = \boldsymbol{H}_t \left(\boldsymbol{x}_t - \hat{\boldsymbol{x}}_t^-\right) + \boldsymbol{v}_t,$$

so that, using the independence of $\boldsymbol{x}_t$ and $\boldsymbol{v}_t$,

$$\boldsymbol{S}_t = \mathrm{Var}(\tilde{\boldsymbol{y}}_t \mid \boldsymbol{y}_{1:t-1}) = \boldsymbol{H}_t \boldsymbol{P}_t^- \boldsymbol{H}_t^\top + \boldsymbol{R}_t, \qquad \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1}) = \boldsymbol{P}_t^- \boldsymbol{H}_t^\top.$$

Consequently, conditional on $\boldsymbol{y}_{1:t-1}$, the state and observation are jointly Gaussian,

$$\begin{pmatrix} \boldsymbol{x}_t \\ \boldsymbol{y}_t \end{pmatrix} \Bigg| \boldsymbol{y}_{1:t-1} \sim \mathcal{N}\left[\begin{pmatrix} \hat{\boldsymbol{x}}_t^- \\ \boldsymbol{H}_t \hat{\boldsymbol{x}}_t^- \end{pmatrix}, \begin{pmatrix} \boldsymbol{P}_t^- & \boldsymbol{P}_t^- \boldsymbol{H}_t^\top \\ \boldsymbol{H}_t \boldsymbol{P}_t^- & \boldsymbol{S}_t \end{pmatrix}\right].$$

Applying the standard formula for the conditional distribution of a jointly Gaussian vector gives

$$\hat{\boldsymbol{x}}_t = \hat{\boldsymbol{x}}_t^- + \boldsymbol{P}_t^- \boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1} \tilde{\boldsymbol{y}}_t, \qquad \boldsymbol{P}_t = \boldsymbol{P}_t^- - \boldsymbol{P}_t^- \boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1} \boldsymbol{H}_t \boldsymbol{P}_t^-.$$

Defining the Kalman gain $\boldsymbol{K}_t = \boldsymbol{P}_t^- \boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1}$ recovers the compact update equations of Section 3.2:

$$\boldsymbol{K}_t = \boldsymbol{P}_t^- \boldsymbol{H}_t^\top \left(\boldsymbol{H}_t \boldsymbol{P}_t^- \boldsymbol{H}_t^\top + \boldsymbol{R}_t\right)^{-1}, \quad \hat{\boldsymbol{x}}_t = \hat{\boldsymbol{x}}_t^- + \boldsymbol{K}_t \tilde{\boldsymbol{y}}_t, \quad \boldsymbol{P}_t = (\boldsymbol{I} - \boldsymbol{K}_t \boldsymbol{H}_t)\, \boldsymbol{P}_t^-.$$